Introduction to APIs and JSONs

INTERMEDIATE IMPORTING DATA IN PYTHON

Hugo Bowne-Anderson
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datacamp

APIs

- Application Programming Interface
- Protocols and routines
  - Building and interacting with software applications

OMDb API

The Open Movie Database

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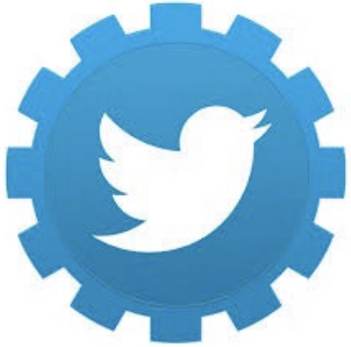
  - Building and interacting with software applications

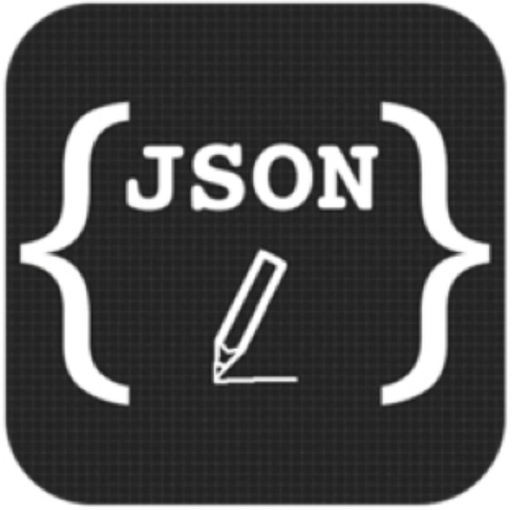
OMDb API

The Open Movie Database



JSONs

- JavaScript Object Notation
- Real-time server-to-browser communication
- Douglas Crockford
- Human readable



JSONs

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{'Actors': 'Samuel L. Jackson, Julianna Margulies, Nathan Phillips,
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 'Awards': '3 wins & 7 nominations.',
 'Country': 'Germany, USA, Canada',
 'Director': 'David R. Ellis',
 'Genre': 'Action, Adventure, Crime',
 'Language': 'English',
 'Rated': 'R',
 'Released': '18 Aug 2006',
 'Runtime': '105 min',
 'Title': 'Snakes on a Plane',
 'Type': 'movie',
 'Writer': 'John Heffernan (screenplay), Sebastian Gutierrez (screenplay),
 David Dalessandro (story), John Heffernan (story)',
 'Year': '2006',
 'imdbID': 'tt0417148',
 'imdbRating': '5.6',
 'imdbVotes': '114,668'}
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Loading JSONs in Python

```
import json
with open('snakes.json', 'r') as json_file:
        json_data = json.load(json_file)
```

```
type(json_data)
```

```
dict
```

Exploring JSONs in Python

```python
for key, value in json_data.items():
    print(key + ':', value)
```

```
Title: Snakes on a Plane
Country: Germany, USA, Canada
Response: True
Language: English
Awards: 3 wins & 7 nominations.
Year: 2006
Actors: Samuel L. Jackson, Julianna Margulies
Runtime: 105 min
Genre: Action, Adventure, Crime
imdbID: tt0417148
Director: David R. Ellis
imdbRating: 5.6
Rated: R
Released: 18 Aug 2006
```

Let's practice!

INTERMEDIATE IMPORTING DATA IN PYTHON

APIs and interacting with the world wide web

INTERMEDIATE IMPORTING DATA IN PYTHON

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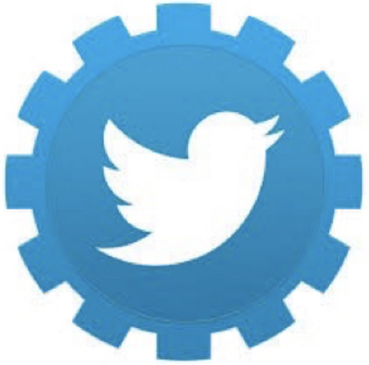
Herein, you'll learn

- What APIs are
- Why APIs are important
- In the exercises:
  - Connecting to APIs
  - Pulling data from APIs
  - Parsing data from APIs

What is an API?

- Set of protocols and routines
- Bunch of code
  - Allows two software programs to communicate with each other



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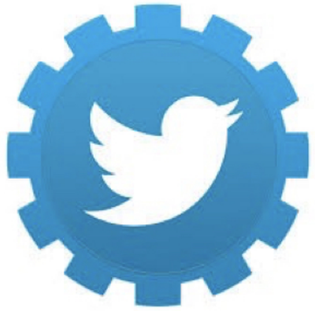
APIs are everywhere



APIs are everywhere

UBER Developers

APIs are everywhere

APIs are everywhere

Connecting to an API in Python

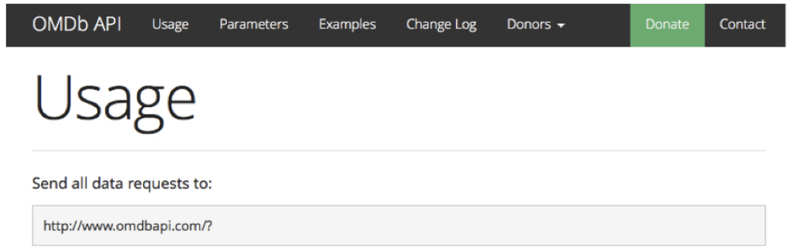
```
import requests
url = 'http://www.omdbapi.com/?t=hackers'
r = requests.get(url)
json_data = r.json()
for key, value in json_data.items():
    print(key + ':', value)
```

What was that URL?

- http - making an HTTP request
- www.omdbapi.com - querying the OMDB API
- `?t=hackers`
  - Query string
  - Return data for a movie with title (t) ‘Hackers’

```
'http://www.omdbapi.com/?t=hackers'
```

OMDb API


OMDb API
Usage
Parameters
Examples
Change Log
Donors
Donate
Contact
Usage
Send all data requests to:
http://www.omdbapi.com/?

OMDb API

OMDb API | Usage | Parameters | Examples | Change Log | Donors | Donate | Contact

Usage

Send all data requests to:

http://www.omdbapi.com/?

Parameters

By ID or Title

Parameter	Required	Valid Options	Default Value	Description
i	Optional*		<empty>	A valid IMDb ID (e.g. tt1285016)
t	Optional*		<empty>	Movie title to search for.
type	No	movie, series, episode	<empty>	Type of result to return.

It's a regular URL!

www.omdbapi.com/?t=hackers

Raw Parsed


```
{
  "Title": "Hackers",
  "Year": "1995",
  "Rated": "PG-13",
  "Released": "15 Sep 1995",
  "Runtime": "107 min",
  "Genre": "Comedy, Crime, Drama",
  "Director": "Iain Softley",
  "Writer": "Rafael Moreu",
  "Actors": "Jonny Lee Miller, Angelina Jolie, Jesse Bradford, Matthew Lillard",
  "Plot": "A young boy is arrested by the U.S. Secret Service for writing a computer virus and is banned from using a computer until his 18th birthday. Years later, he and his new-found friends ...",
  "Language": "English, Italian, Russian, Japanese",
  "Country": "USA",
  "Awards": "N/A",
  "Poster": "http://ia.media-imdb.com/images/M/MV5BODg0NjQ5ODQ3OF5BMl5BanBnXkFtZTcwNjU4MjkzNA@@._V1_SX300.jpg",
  "Metascore": "46",
  "imdbRating": "6.2",
  "imdbVotes": "53,874",
  "imdbID": "tt0113243",
  "Type": "movie",
  "Response": "True"
}
```

Let's practice!

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